Nicholas Steffen

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Software developer with 10+ years of experience in full-stack web development, integrations, and internal tooling in SaaS environments. Skilled in API integrations, automated testing, and leveraging open-source tools. Experienced in version control best practices and collaborative workflows. Adaptable and driven by a growth mindset with strong communication skills, a focus on problem-solving, and a commitment to working effectively with cross-functional teams.

Skills

Languages & Frameworks: Python, Java, C++, Typescript, Node.js, FastAPI, Flask, Next.js, React, React Native, Vue.js
Data Technologies: MySQL, PostgreSQL, MongoDB, SQLite, Redis, ElasticSearch, RabbitMQ, NoSQL, SQL
Cloud & DevOps: Docker, Linux/Unix, Bash, Ansible, Google Cloud Platform, AWS
Tools and APIs: REST, JIRA, JUnit, Maven, JWT, OAuth, Supabase, Vercel
Professional Skills & Methodologies: Agile, UI/UX Design, Open Source, Problem-solving, Communications, Teamwork, critical thinking, SaaS product, Distributed Systems

Professional Experience

Independent Consultant **December 2025 - Present**
Lead Full-Stack Consultant *Remote*

- Architected a scalable, serverless backend using Supabase Edge Functions and PostgreSQL, migrating from a FastAPI prototype to optimize for performance and cost.
- Designed and implemented a custom CI/CD pipeline on Vercel with a multi-branch environment (Main, Develop, Staging, Production) to safely vet AI-assisted code contributions.
- Developed and maintained a high-performance Vite web application with feature parity and seamless data synchronization with the native iOS platform.
- Engineered core platform features including secure Supabase Authentication and a cross-platform Stripe payment integration.
- Directed technical launch operations, establishing health monitoring systems and Vercel Analytics to track performance during high-traffic events.
- Collaborated with iOS engineers to align database schema designs and backend infrastructure with mobile-specific constraints.

ClearlyRated **June 2021 - November 2024**
Software Engineer *Remote*

- Spearheaded a critical backend modernization initiative, researching and implementing the Quart framework for async capabilities (from Flask), achieving a minimally disruptive transition while also identifying and resolving a long-standing logging issue and configuring Hypercorn for robust deployment.
- Architected and developed a robust, agnostic Java object model for third-party API integrations, accelerating the onboarding of new CRM platforms (e.g., Salesforce, Microsoft Dynamics) by enabling direct integration with existing mapping and filtering tools. This reusable architecture empowered other developers to rapidly build new UIs and integrate diverse data sources efficiently.
- Developed internal developer tools to containerize MySQL databases using Docker, leveraging Python scripting to produce and populate reproducible, large-scale, production-like data. This initiative aimed to streamline feature development, accelerate environment setup, and allow for early detection of scaling issues by providing robust, consistent data. It also laid foundational support for enhanced CI/CD pipeline testing and QA processes.

- Led a research initiative to enhance unit-test efficiency, identifying bottlenecks caused by long-running tests that impeded CI/CD pipelines and developer productivity. Proposed and developed a strategic plan leveraging JUnit flows to isolate and refactor tests using mocks, laying the groundwork for accelerated testing cycles and streamlined development workflows.
- Engineered a full-stack solution (Vue.js, Java backend, MySQL) enabling users to upload, validate, and process CSV contact data, streamlining customer data ingestion and expanding integration capabilities by abstracting existing backend logic for broader applicability.

Raytheon **August 2018 - May 2021**
Software Engineer II *Portsmouth, RI, USA*

- Architected and developed a robust device network monitoring suite using Python and SNMP, integrating with legacy Java services via Redis to enable seamless communication. This critical system was designed to proactively monitor hardware health, automatically issuing shutdown signals to prevent overheating and safeguard expensive equipment, thereby ensuring operational continuity and significantly reducing potential damage costs.
- Led a research initiative to enhance system observability and reliability for mission-critical hardware, evaluating log monitoring solutions against a strict 99% uptime requirement. Recommended and implemented Splunk, designing a highly available architecture with dual instances and failover capabilities to ensure continuous data capture.
- Developed Python API scripts to ensure redundant data transmission to both Splunk instances, minimizing data loss. Additionally, authored Ansible playbooks and Kickstart files for automated deployment and configuration across a fleet of hardware devices and Splunk servers, streamlining setup and maintaining configuration consistency.

Computer Associates Inc **June 2016 - July 2018**
Software Consultant *Smithfield, RI, USA*

- Collaborated directly with diverse clients as a Software Consultant, gathering requirements and advising on solutions to customize core ERP software. Led the development of client-specific web applications, including a custom hardwood door configurator, enabling customers to design products, generate quotes, and access sales data online, significantly enhancing client self-service and expanding business reach

Farsounder **May 2015 - May 2016**
Software Engineer Intern *Warwick, RI, USA*

- Implemented improvements to C++ based map overlays to track GPS position and provide real-time interactive distance and angle calculations (leveraging trigonometry) from the ship's heading to a cursor-defined point, enhancing user spatial awareness and navigation capabilities.
- Developed an enhancement to an internal data playback tool by implementing tag support and a hash-based dataset lookup, significantly streamlining developer workflows and increasing productivity by enabling rapid retrieval of specific playback file types for efficient local testing of new features
- Led a research project investigating the feasibility of forward-looking sonar for depth approximation, with the objective of enhancing ship safety and enabling real-time ocean floor mapping for clients. Designed a methodology to validate predictions against existing down-looking depth finders, laying foundational insights for potential future navigation and mapping product development.

Education

Providence College **September 2012 - May 2016**
Bachelor's, Applied Mathematics

- Minor: Computer Science